Students' Development of Statistical Narratives in Probability Modeling SEM

Issues So Far

- Total N = **84**
- Complete cases = **40**
- A lot of missing observations in NE4 (30 missing) and NE5 (27 missing), casuing the full SEM to become unstable due to small sample size

```r
library(lavaan) # for SEM
```

```
## This is lavaan 0.6-21
## lavaan is FREE software! Please report any bugs.
```

```r
library(tidyverse, quietly = TRUE) # for data set join, file link read
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
# get data from all three activities
A3 = read_csv("https://raw.githubusercontent.com/jfleege/narrative-sem/main/datasets/CombinedActivity3.
  rename(Anonymous.ID = `Anonymous ID`)
```

```
## New names:
## * 'If your sampler functionally differed from the one you chose above, describe
##   how.' -> 'If your sampler functionally differed from the one you chose above,
##   describe how....6'
## * 'If your sampler functionally differed from the one you chose above, describe
##   how.' -> 'If your sampler functionally differed from the one you chose above,
##   describe how....9'
## * 'Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?' -> 'Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...10'
## * 'Describe how talking about this device in class did (or didn't) support your
##   learning in class.' -> 'Describe how talking about this device in class did
##   (or didn't) support your learning in class....11'
## * 'Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?' -> 'Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...12'
## * 'Describe how talking about this device in class did (or didn't) support your
##   learning in class.' -> 'Describe how talking about this device in class did
```

```
##   (or didn't) support your learning in class....13`
## * `Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?` -> `Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...14`
## * `Describe how talking about this device in class did (or didn't) support your
##   learning in class.` -> `Describe how talking about this device in class did
##   (or didn't) support your learning in class....15`
```

```
A3$Survey_NE = as.numeric(A3$Survey_NE)
```

```
## Warning: NAs introduced by coercion
```

```
A4 = read_csv("https://raw.githubusercontent.com/jfleege/narrative-sem/main/datasets/CombinedActivity4.c
  rename(Anonymous.ID = `Anonymous ID`)

A4$Survey_NE = as.numeric(A4$Survey_NE)

A5 = read_csv("https://raw.githubusercontent.com/jfleege/narrative-sem/main/datasets/CombinedActivity5.c
  rename(Anonymous.ID = `Anonymous ID`)
```

```
## New names:
## * `If your sampler functionally differed from the one you chose above, describe
##   how.` -> `If your sampler functionally differed from the one you chose above,
##   describe how....6`
## * `If your sampler functionally differed from the one you chose above, describe
##   how.` -> `If your sampler functionally differed from the one you chose above,
##   describe how....9`
## * `Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?` -> `Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...10`
## * `Describe how talking about this device in class either did, or didn't,
##   support your learning in class.` -> `Describe how talking about this device
##   in class either did, or didn't, support your learning in class....11`
## * `Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?` -> `Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...12`
## * `Describe how talking about this device in class either did, or didn't,
##   support your learning in class.` -> `Describe how talking about this device
##   in class either did, or didn't, support your learning in class....13`
## * `Describe how talking about this device in class did (or didn't) support your
##   learning in class.` -> `Describe how talking about this device in class did
##   (or didn't) support your learning in class....15`
## * `Upon observation of the given device, what about this sampler appeals to
##   you? What doesn't appeal to you?` -> `Upon observation of the given device,
##   what about this sampler appeals to you? What doesn't appeal to you?...16`
## * `Describe how talking about this device in class did (or didn't) support your
##   learning in class.` -> `Describe how talking about this device in class did
##   (or didn't) support your learning in class....17`
```

```
A5$Survey_NE = as.numeric(A5$Survey_NE)
```

```
## Warning: NAs introduced by coercion
```

```r
# establish ANIQ columns
item_cols <- c("AW1","AW2","AW3","AW4","AW5",
               "TM1","TM2","TM3","TM4","TM5",
               "CA1","CA2","CA3","CA4","CA5",
               "TH1","TH2","TH3","TH4","TH5")

# combine data across all three activities
SEMcombined <- A3 %>%
  select(Anonymous.ID, all_of(item_cols), NE3 = Survey_NE) %>%
  left_join(
    A4 %>% select(Anonymous.ID, NE4 = Survey_NE),
    by = "Anonymous.ID"
  ) %>%
  left_join(
    A5 %>% select(Anonymous.ID, NE5 = Survey_NE),
    by = "Anonymous.ID"
  )

# convert ANIQ item columns to numeric
SEMcombined[item_cols] <- lapply(
  SEMcombined[item_cols],
  function(x) as.numeric(as.character(x))
)
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion

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## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```r
# order data
SEMcombined$NE3 <- as.ordered(SEMcombined$NE3)
SEMcombined$NE4 <- as.ordered(SEMcombined$NE4)
SEMcombined$NE5 <- as.ordered(SEMcombined$NE5)
```

Data Set Information

```r
sum(complete.cases(SEMcombined)) # complete cases
```

```
## [1] 40
```

```r
nrow(SEMcombined) # total number of observations (students)
```

```
## [1] 84
```

```r
colSums(is.na(SEMcombined)) # number of missing values per column
```

```
## Anonymous.ID          AW1          AW2          AW3          AW4          AW5
##            0            7            7            7            7            7
##          TM1          TM2          TM3          TM4          TM5          CA1
##            7            7            7            7            7            7
##          CA2          CA3          CA4          CA5          TH1          TH2
##            7            7            7            7            7            7
##          TH3          TH4          TH5          NE3          NE4          NE5
##            7            7            7            1           30           27
```

I thought it might be helpful to include this brief information about the data set before diving into the models to get a better understanding of why we might be encountering certain problems. After running these commands, I found that there are 40 complete cases out of 84 total observations (probably not good), with some missing values in the ANIQ item columns.

Model 1: Baseline

This was the first structure made and tested, with no changes to any of the ANIQ columns or subcategories. Not favored due to the bad scaled CFI, RMSEA, and SRMR values of 0.622, 0.549, and 0.123, respectively. Initial conclusion for the bad fit was due to the lack of overall used observations (n = 40) given the amount of ANIQ parameters.

```r
model1 <- "
  aw =~ AW1 + AW2 + AW3 + AW4 + AW5
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA1 + CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH2 + TH3 + TH4 + TH5

  NE3 ~ aw + tm + ca + th
  NE4 ~ aw + tm + ca + th
  NE5 ~ aw + tm + ca + th
"

fit1 <- sem(model1, data = SEMcombined, ordered = c("NE3", "NE4", "NE5"))
```

```
## Warning: lavaan->lav_model_vcov():
##    The variance-covariance matrix of the estimated parameters (vcov) does not
##    appear to be positive definite! The smallest eigenvalue (= -1.067524e-11)
##    is smaller than zero. This may be a symptom that the model is not
##    identified.
```

```
## Warning: lavaan->lav_object_post_check():
##    some estimated ov variances are negative
```

```r
summary(fit1, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 149 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                          84
##
##                                                   Used       Total
##   Number of observations                            40          84
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               142.274     301.016
##   Degrees of freedom                               212         212
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.979
##   Shift parameter                                          155.684
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              1601.171     488.360
##   Degrees of freedom                               253         253
##   P-value                                           NA       0.000
##   Scaling correction factor                                  5.728
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.622
##   Tucker-Lewis Index (TLI)                       1.062       0.549
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.104
##   90 Percent confidence interval - lower         0.000       0.075
##   90 Percent confidence interval - upper         0.000       0.130
##   P-value H_0: RMSEA <= 0.050                    1.000       0.002
##   P-value H_0: RMSEA >= 0.080                    0.000       0.920
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
```

```
##   SRMR                                           0.123       0.123
##
## Parameter Estimates:
##
##   Parameterization                               Delta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.655    0.886
##     AW2               0.879    0.183    4.809    0.000    1.455    0.762
##     AW3               0.469    0.137    3.426    0.001    0.776    0.484
##     AW4               0.929    0.215    4.330    0.000    1.538    0.706
##     AW5               1.020    0.277    3.688    0.000    1.688    0.847
##   tm =~
##     TM1               1.000                               2.079    0.946
##     TM2               0.994    0.131    7.571    0.000    2.067    0.975
##     TM3               0.655    0.105    6.235    0.000    1.362    0.665
##     TM4               0.770    0.098    7.870    0.000    1.601    0.878
##     TM5               0.880    0.133    6.632    0.000    1.829    0.917
##   ca =~
##     CA1               1.000                               1.255    0.634
##     CA2               1.164    0.314    3.711    0.000    1.461    0.706
##     CA3               0.951    0.256    3.714    0.000    1.193    0.601
##     CA4               1.040    0.240    4.337    0.000    1.305    0.794
##     CA5               0.935    0.267    3.506    0.000    1.173    0.662
##   th =~
##     TH1               1.000                               0.860    0.567
##     TH2               1.868    0.535    3.494    0.000    1.606    0.830
##     TH3               1.520    0.313    4.862    0.000    1.307    0.737
##     TH4               1.455    0.412    3.532    0.000    1.251    0.732
##     TH5               1.841    0.378    4.868    0.000    1.583    0.822
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~
##     aw                1.310    0.860    1.524    0.128    2.168    2.168
##     tm               -0.157    0.563   -0.278    0.781   -0.326   -0.326
##     ca                1.121    1.209    0.927    0.354    1.408    1.408
##     th               -4.224    2.133   -1.980    0.048   -3.632   -3.632
##   NE4 ~
##     aw               -1.036    0.494   -2.099    0.036   -1.715   -1.715
##     tm                0.126    0.320    0.395    0.693    0.262    0.262
##     ca               -0.818    0.859   -0.952    0.341   -1.027   -1.027
##     th                2.898    1.262    2.296    0.022    2.492    2.492
##   NE5 ~
##     aw               -0.128    0.390   -0.329    0.742   -0.212   -0.212
##     tm                0.160    0.202    0.792    0.428    0.333    0.333
##     ca               -0.104    0.521   -0.200    0.841   -0.131   -0.131
##     th                0.397    0.794    0.500    0.617    0.342    0.342
##
```

```
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     tm                1.722    0.650    2.648    0.008    0.501    0.501
##     ca                1.566    0.581    2.695    0.007    0.754    0.754
##     th                1.298    0.403    3.222    0.001    0.912    0.912
##   tm ~~
##     ca                2.037    0.789    2.581    0.010    0.781    0.781
##     th                0.928    0.353    2.629    0.009    0.519    0.519
##   ca ~~
##     th                0.885    0.340    2.601    0.009    0.820    0.820
##  .NE3 ~~
##    .NE4               0.709    0.575    1.232    0.218    0.709    1.756
##    .NE5               0.313    0.282    1.108    0.268    0.313    0.416
##  .NE4 ~~
##    .NE5              -0.247    0.291   -0.849    0.396   -0.247   -0.514
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.250    0.348   20.815    0.000    7.250    3.882
##    .AW2               7.175    0.344   20.858    0.000    7.175    3.758
##    .AW3               8.150    0.317   25.725    0.000    8.150    5.076
##    .AW4               6.900    0.388   17.765    0.000    6.900    3.169
##    .AW5               6.925    0.346   20.039    0.000    6.925    3.476
##    .TM1               6.850    0.406   16.854    0.000    6.850    3.118
##    .TM2               7.100    0.441   16.088    0.000    7.100    3.351
##    .TM3               7.525    0.435   17.298    0.000    7.525    3.672
##    .TM4               7.025    0.410   17.128    0.000    7.025    3.853
##    .TM5               6.650    0.367   18.135    0.000    6.650    3.334
##    .CA1               6.975    0.349   19.957    0.000    6.975    3.521
##    .CA2               7.150    0.392   18.261    0.000    7.150    3.457
##    .CA3               7.050    0.349   20.176    0.000    7.050    3.548
##    .CA4               7.500    0.329   22.820    0.000    7.500    4.564
##    .CA5               7.425    0.303   24.504    0.000    7.425    4.187
##    .TH1               7.500    0.274   27.405    0.000    7.500    4.945
##    .TH2               7.175    0.335   21.426    0.000    7.175    3.708
##    .TH3               7.425    0.322   23.075    0.000    7.425    4.187
##    .TH4               7.325    0.312   23.469    0.000    7.325    4.287
##    .TH5               7.125    0.331   21.523    0.000    7.125    3.699
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3|t1           -1.036    0.245   -4.226    0.000   -1.036   -1.036
##     NE4|t1           -0.385    0.206   -1.869    0.062   -0.385   -0.385
##     NE5|t1            0.189    0.202    0.936    0.349    0.189    0.189
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.749    0.350    2.139    0.032    0.749    0.215
##    .AW2               1.529    0.412    3.714    0.000    1.529    0.419
##    .AW3               1.975    0.315    6.263    0.000    1.975    0.766
##    .AW4               2.374    0.573    4.142    0.000    2.374    0.501
##    .AW5               1.119    0.326    3.429    0.001    1.119    0.282
##    .TM1               0.506    0.428    1.183    0.237    0.506    0.105
```

```
##    .TM2               0.220    0.366    0.600    0.549    0.220    0.049
##    .TM3               2.343    0.320    7.322    0.000    2.343    0.558
##    .TM4               0.761    0.299    2.545    0.011    0.761    0.229
##    .TM5               0.632    0.412    1.533    0.125    0.632    0.159
##    .CA1               2.349    0.496    4.739    0.000    2.349    0.599
##    .CA2               2.143    0.299    7.164    0.000    2.143    0.501
##    .CA3               2.523    0.302    8.356    0.000    2.523    0.639
##    .CA4               0.996    0.178    5.598    0.000    0.996    0.369
##    .CA5               1.768    0.301    5.870    0.000    1.768    0.562
##    .TH1               1.561    0.201    7.782    0.000    1.561    0.679
##    .TH2               1.164    0.317    3.669    0.000    1.164    0.311
##    .TH3               1.436    0.256    5.604    0.000    1.436    0.457
##    .TH4               1.355    0.377    3.599    0.000    1.355    0.464
##    .TH5               1.203    0.332    3.627    0.000    1.203    0.324
##    .NE3              -0.632                              -0.632   -0.632
##    .NE4               0.258                               0.258    0.258
##    .NE5               0.895                               0.895    0.895
##     aw                2.738    1.054    2.598    0.009    1.000    1.000
##     tm                4.321    1.516    2.851    0.004    1.000    1.000
##     ca                1.576    0.753    2.092    0.036    1.000    1.000
##     th                0.739    0.300    2.461    0.014    1.000    1.000
```

Model 2: EFA Adjustments

This model was similar to the first; however, removed the ANIQ items that either had low factor loadings or cross-loadings based on Jon's findings. Based on these adjustments, noticed that CFI improved slightly to 0.662 and TLI improved to 0.577, but SRMR fell to 0.110 on n = 40 observations.

```r
model2 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE3 ~ aw + tm + ca + th
  NE4 ~ aw + tm + ca + th
  NE5 ~ aw + tm + ca + th
"

fit2 <- sem(model2, data = SEMcombined, ordered = c("NE3", "NE4", "NE5"))
```

```
## Warning: lavaan->lav_model_vcov():
##    The variance-covariance matrix of the estimated parameters (vcov) does not
##    appear to be positive definite! The smallest eigenvalue (= -1.624806e-13)
##    is smaller than zero. This may be a symptom that the model is not
##    identified.

## Warning: lavaan->lav_object_post_check():
##    some estimated ov variances are negative
```

```r
summary(fit2, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 102 iterations
##
##   Estimator                                         DWLS
##   Optimization method                             NLMINB
##   Number of model parameters                          75
##
##                                                   Used       Total
##   Number of observations                            40          84
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               106.171     223.651
##   Degrees of freedom                               152         152
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.898
##   Shift parameter                                          105.363
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              1210.830     401.741
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.821
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.662
##   Tucker-Lewis Index (TLI)                       1.056       0.577
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.110
##   90 Percent confidence interval - lower         0.000       0.078
##   90 Percent confidence interval - upper         0.000       0.140
##   P-value H_0: RMSEA <= 0.050                    1.000       0.003
##   P-value H_0: RMSEA >= 0.080                    0.000       0.937
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.121       0.121
##
## Parameter Estimates:
##
##   Parameterization                               Delta
```

```
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.769    0.947
##     AW2               0.828    0.178    4.665    0.000    1.466    0.768
##     AW3               0.475    0.141    3.380    0.001    0.841    0.524
##     AW4               0.929    0.227    4.098    0.000    1.643    0.755
##   tm =~
##     TM1               1.000                               2.082    0.948
##     TM2               0.988    0.131    7.536    0.000    2.058    0.971
##     TM3               0.656    0.108    6.050    0.000    1.365    0.666
##     TM4               0.762    0.098    7.738    0.000    1.587    0.870
##     TM5               0.887    0.133    6.648    0.000    1.846    0.926
##   ca =~
##     CA2               1.000                               1.497    0.724
##     CA3               0.856    0.108    7.934    0.000    1.281    0.645
##     CA4               0.889    0.226    3.933    0.000    1.330    0.809
##     CA5               0.771    0.188    4.103    0.000    1.154    0.651
##   th =~
##     TH1               1.000                               0.881    0.581
##     TH3               1.553    0.331    4.696    0.000    1.368    0.771
##     TH4               1.512    0.432    3.499    0.000    1.331    0.779
##     TH5               1.830    0.394    4.650    0.000    1.612    0.837
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~
##     aw                0.572    0.253    2.266    0.023    1.013    1.013
##     tm                0.021    0.297    0.070    0.944    0.043    0.043
##     ca                0.407    0.458    0.888    0.375    0.608    0.608
##     th               -2.342    0.716   -3.271    0.001   -2.063   -2.063
##   NE4 ~
##     aw               -0.463    0.162   -2.854    0.004   -0.819   -0.819
##     tm               -0.071    0.124   -0.571    0.568   -0.147   -0.147
##     ca               -0.080    0.318   -0.250    0.803   -0.119   -0.119
##     th                1.232    0.361    3.418    0.001    1.085    1.085
##   NE5 ~
##     aw                0.052    0.210    0.249    0.804    0.092    0.092
##     tm                0.087    0.164    0.531    0.595    0.181    0.181
##     ca                0.159    0.324    0.492    0.623    0.239    0.239
##     th               -0.227    0.345   -0.658    0.510   -0.200   -0.200
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     tm                1.806    0.667    2.708    0.007    0.490    0.490
##     ca                1.917    0.725    2.644    0.008    0.724    0.724
##     th                1.233    0.399    3.090    0.002    0.791    0.791
##   tm ~~
##     ca                2.269    0.777    2.921    0.003    0.728    0.728
```

```
##     th                0.897    0.349    2.570    0.010    0.489    0.489
##   ca ~~
##     th                0.976    0.331    2.949    0.003    0.741    0.741
##  .NE3 ~~
##    .NE4               0.361    0.302    1.195    0.232    0.361    0.773
##    .NE5               0.051    0.255    0.201    0.841    0.051    0.089
##  .NE4 ~~
##    .NE5              -0.090    0.237   -0.379    0.704   -0.090   -0.125
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.250    0.348   20.815    0.000    7.250    3.882
##    .AW2               7.175    0.344   20.858    0.000    7.175    3.758
##    .AW3               8.150    0.317   25.725    0.000    8.150    5.076
##    .AW4               6.900    0.388   17.765    0.000    6.900    3.169
##    .TM1               6.850    0.406   16.854    0.000    6.850    3.118
##    .TM2               7.100    0.441   16.088    0.000    7.100    3.351
##    .TM3               7.525    0.435   17.298    0.000    7.525    3.672
##    .TM4               7.025    0.410   17.128    0.000    7.025    3.853
##    .TM5               6.650    0.367   18.135    0.000    6.650    3.334
##    .CA2               7.150    0.392   18.261    0.000    7.150    3.457
##    .CA3               7.050    0.349   20.176    0.000    7.050    3.548
##    .CA4               7.500    0.329   22.820    0.000    7.500    4.564
##    .CA5               7.425    0.303   24.504    0.000    7.425    4.187
##    .TH1               7.500    0.274   27.405    0.000    7.500    4.945
##    .TH3               7.425    0.322   23.075    0.000    7.425    4.187
##    .TH4               7.325    0.312   23.469    0.000    7.325    4.287
##    .TH5               7.125    0.331   21.523    0.000    7.125    3.699
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3|t1           -1.036    0.245   -4.226    0.000   -1.036   -1.036
##     NE4|t1           -0.385    0.206   -1.869    0.062   -0.385   -0.385
##     NE5|t1            0.189    0.202    0.936    0.349    0.189    0.189
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.357    0.491    0.727    0.467    0.357    0.102
##    .AW2               1.496    0.480    3.113    0.002    1.496    0.410
##    .AW3               1.870    0.317    5.903    0.000    1.870    0.726
##    .AW4               2.041    0.608    3.356    0.001    2.041    0.431
##    .TM1               0.491    0.419    1.174    0.241    0.491    0.102
##    .TM2               0.255    0.342    0.746    0.456    0.255    0.057
##    .TM3               2.336    0.319    7.331    0.000    2.336    0.556
##    .TM4               0.807    0.286    2.817    0.005    0.807    0.243
##    .TM5               0.570    0.392    1.455    0.146    0.570    0.143
##    .CA2               2.037    0.316    6.455    0.000    2.037    0.476
##    .CA3               2.305    0.278    8.295    0.000    2.305    0.584
##    .CA4               0.931    0.174    5.345    0.000    0.931    0.345
##    .CA5               1.812    0.339    5.354    0.000    1.812    0.576
##    .TH1               1.524    0.194    7.845    0.000    1.524    0.663
##    .TH3               1.274    0.278    4.589    0.000    1.274    0.405
##    .TH4               1.147    0.404    2.835    0.005    1.147    0.393
##    .TH5               1.112    0.361    3.080    0.002    1.112    0.300
```

```
##    .NE3              -0.374                              -0.374   -0.374
##    .NE4               0.585                               0.585    0.585
##    .NE5               0.886                               0.886    0.886
##     aw                3.130    1.173    2.668    0.008    1.000    1.000
##     tm                4.336    1.538    2.819    0.005    1.000    1.000
##     ca                2.240    0.967    2.316    0.021    1.000    1.000
##     th                0.776    0.319    2.432    0.015    1.000    1.000
```

Model 3: Activity Dependency

This model was made considering that the latter activities (4 and 5) were dependent on the previous as the students progressed through the activities. Additionally, this model uses the same subcategories as Model 2. This model had the best fit of the three with a CFI of 0.967, TLI of 0.959, and SRMR of 0.082. Unsure if this is due to over fitting.

```r
model3 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE3 ~ aw + tm + ca + th
  NE4 ~ NE3 + aw + tm + ca + th
  NE5 ~ NE4 + aw + tm + ca + th
"

fit3 <- sem(
  model3,
  data = SEMcombined,
  ordered = c(item_cols, "NE3","NE4","NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_model_vcov():
##    The variance-covariance matrix of the estimated parameters (vcov) does not
##    appear to be positive definite! The smallest eigenvalue (= -1.187517e-12)
##    is smaller than zero. This may be a symptom that the model is not
##    identified.
```

```r
summary(fit3, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 204 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                       171
##
##   Number of observations                            84
##   Number of missing patterns                         8
##
```

```
## Model Test User Model:
##                                              Standard      Scaled
##   Test Statistic                              184.064     256.932
##   Degrees of freedom                              153         153
##   P-value (Unknown)                                NA       0.000
##   Scaling correction factor                                 1.069
##   Shift parameter                                          84.804
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                            10371.752    3331.610
##   Degrees of freedom                              190         190
##   P-value                                          NA       0.000
##   Scaling correction factor                                 3.241
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                   0.997       0.967
##   Tucker-Lewis Index (TLI)                      0.996       0.959
##
##   Robust Comparative Fit Index (CFI)                           NA
##   Robust Tucker-Lewis Index (TLI)                              NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                         0.049       0.090
##   90 Percent confidence interval - lower        0.009       0.071
##   90 Percent confidence interval - upper        0.074       0.109
##   P-value H_0: RMSEA <= 0.050                   0.498       0.001
##   P-value H_0: RMSEA >= 0.080                   0.017       0.818
##
##   Robust RMSEA                                                 NA
##   90 Percent confidence interval - lower                       NA
##   90 Percent confidence interval - upper                       NA
##   P-value H_0: Robust RMSEA <= 0.050                           NA
##   P-value H_0: Robust RMSEA >= 0.080                           NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                          0.082       0.082
##
## Parameter Estimates:
##
##   Parameterization                              Theta
##   Standard errors                          Robust.sem
##   Information                                Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               2.620    0.934
##     AW2               0.412    0.123    3.352    0.001    1.080    0.734
```

```
##     AW3               0.299    0.081    3.691    0.000    0.782    0.616
##     AW4               0.569    0.165    3.453    0.001    1.491    0.830
##   tm =~
##     TM1               1.000                               2.322    0.918
##     TM2               1.029    0.273    3.767    0.000    2.390    0.923
##     TM3               0.454    0.099    4.590    0.000    1.053    0.725
##     TM4               0.775    0.162    4.790    0.000    1.799    0.874
##     TM5               0.738    0.174    4.237    0.000    1.713    0.864
##   ca =~
##     CA2               1.000                               1.507    0.833
##     CA3               0.790    0.123    6.444    0.000    1.190    0.766
##     CA4               0.943    0.196    4.815    0.000    1.421    0.818
##     CA5               0.749    0.122    6.140    0.000    1.129    0.749
##   th =~
##     TH1               1.000                               0.896    0.667
##     TH3               1.457    0.338    4.311    0.000    1.306    0.794
##     TH4               1.074    0.248    4.337    0.000    0.963    0.694
##     TH5               1.882    0.508    3.702    0.000    1.687    0.860
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~
##     aw                0.738    0.705    1.047    0.295    1.934    1.054
##     tm                0.441    0.445    0.991    0.322    1.024    0.558
##     ca               -0.566    0.703   -0.805    0.421   -0.853   -0.465
##     th               -2.805    2.480   -1.131    0.258   -2.514   -1.371
##   NE4 ~
##     NE3              -0.156    0.552   -0.283    0.777   -0.156   -0.214
##     aw               -0.562    0.631   -0.890    0.374   -1.471   -1.097
##     tm               -0.003    0.267   -0.010    0.992   -0.006   -0.004
##     ca                0.015    0.537    0.028    0.977    0.023    0.017
##     th                1.379    1.957    0.705    0.481    1.236    0.922
##   NE5 ~
##     NE4               0.009    0.345    0.026    0.979    0.009    0.011
##     aw                0.035    0.353    0.100    0.920    0.092    0.086
##     tm                0.081    0.164    0.496    0.620    0.189    0.175
##     ca                0.251    0.349    0.718    0.473    0.378    0.350
##     th               -0.387    0.992   -0.391    0.696   -0.347   -0.321
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     tm                4.010    1.254    3.198    0.001    0.659    0.659
##     ca                3.083    1.082    2.849    0.004    0.781    0.781
##     th                2.063    0.671    3.073    0.002    0.879    0.879
##   tm ~~
##     ca                2.738    0.689    3.974    0.000    0.782    0.782
##     th                1.141    0.330    3.460    0.001    0.548    0.548
##   ca ~~
##     th                1.017    0.259    3.931    0.000    0.752    0.752
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     AW1|t1           -6.243    1.452   -4.300    0.000   -6.243   -2.227
```

```
##     AW1|t2          -4.560    1.052   -4.336    0.000   -4.560   -1.626
##     AW1|t3          -4.247    0.950   -4.473    0.000   -4.247   -1.515
##     AW1|t4          -2.837    0.718   -3.949    0.000   -2.837   -1.012
##     AW1|t5          -1.806    0.524   -3.450    0.001   -1.806   -0.644
##     AW1|t6          -0.228    0.408   -0.560    0.576   -0.228   -0.081
##     AW1|t7           2.037    0.565    3.608    0.000    2.037    0.726
##     AW1|t8           3.744    0.850    4.404    0.000    3.744    1.335
##     AW2|t1          -3.278    0.579   -5.661    0.000   -3.278   -2.227
##     AW2|t2          -2.230    0.333   -6.694    0.000   -2.230   -1.515
##     AW2|t3          -2.089    0.314   -6.661    0.000   -2.089   -1.419
##     AW2|t4          -1.337    0.248   -5.397    0.000   -1.337   -0.908
##     AW2|t5          -0.669    0.219   -3.057    0.002   -0.669   -0.455
##     AW2|t6           0.072    0.212    0.339    0.734    0.072    0.049
##     AW2|t7           0.948    0.226    4.205    0.000    0.948    0.644
##     AW2|t8           1.572    0.258    6.086    0.000    1.572    1.068
##     AW3|t1          -2.827    0.499   -5.670    0.000   -2.827   -2.227
##     AW3|t2          -2.238    0.334   -6.699    0.000   -2.238   -1.763
##     AW3|t3          -1.695    0.254   -6.687    0.000   -1.695   -1.335
##     AW3|t4          -1.356    0.226   -6.003    0.000   -1.356   -1.068
##     AW3|t5          -0.487    0.192   -2.542    0.011   -0.487   -0.384
##     AW3|t6           0.399    0.179    2.233    0.026    0.399    0.314
##     AW3|t7           1.034    0.184    5.613    0.000    1.034    0.814
##     AW4|t1          -3.997    0.716   -5.579    0.000   -3.997   -2.227
##     AW4|t2          -3.489    0.535   -6.522    0.000   -3.489   -1.944
##     AW4|t3          -2.547    0.420   -6.063    0.000   -2.547   -1.419
##     AW4|t4          -1.381    0.307   -4.497    0.000   -1.381   -0.770
##     AW4|t5          -0.688    0.274   -2.511    0.012   -0.688   -0.384
##     AW4|t6           0.323    0.257    1.258    0.208    0.323    0.180
##     AW4|t7           1.229    0.274    4.483    0.000    1.229    0.685
##     AW4|t8           2.261    0.374    6.045    0.000    2.261    1.260
##     TM1|t1          -5.629    1.115   -5.048    0.000   -5.629   -2.227
##     TM1|t2          -3.829    0.549   -6.975    0.000   -3.829   -1.515
##     TM1|t3          -3.376    0.545   -6.197    0.000   -3.376   -1.335
##     TM1|t4          -2.058    0.401   -5.136    0.000   -2.058   -0.814
##     TM1|t5          -1.150    0.368   -3.127    0.002   -1.150   -0.455
##     TM1|t6           0.372    0.364    1.022    0.307    0.372    0.147
##     TM1|t7           1.731    0.450    3.848    0.000    1.731    0.685
##     TM1|t8           3.010    0.564    5.342    0.000    3.010    1.191
##     TM2|t1          -5.035    0.811   -6.209    0.000   -5.035   -1.944
##     TM2|t2          -4.567    0.761   -5.999    0.000   -4.567   -1.763
##     TM2|t3          -4.213    0.731   -5.766    0.000   -4.213   -1.626
##     TM2|t4          -3.085    0.535   -5.770    0.000   -3.085   -1.191
##     TM2|t5          -2.109    0.422   -4.996    0.000   -2.109   -0.814
##     TM2|t6          -0.904    0.376   -2.402    0.016   -0.904   -0.349
##     TM2|t7           0.466    0.383    1.219    0.223    0.466    0.180
##     TM2|t8           1.994    0.479    4.158    0.000    1.994    0.770
##     TM2|t9           3.459    0.576    6.002    0.000    3.459    1.335
##     TM3|t1          -2.823    0.460   -6.141    0.000   -2.823   -1.944
##     TM3|t2          -2.560    0.396   -6.457    0.000   -2.560   -1.763
##     TM3|t3          -2.362    0.368   -6.423    0.000   -2.362   -1.626
##     TM3|t4          -1.729    0.284   -6.097    0.000   -1.729   -1.191
##     TM3|t5          -0.822    0.223   -3.691    0.000   -0.822   -0.566
##     TM3|t6          -0.310    0.211   -1.465    0.143   -0.310   -0.213
##     TM3|t7           0.557    0.207    2.694    0.007    0.557    0.384
```

```
##     TM3|t8          1.550   0.189   8.182   0.000   1.550   1.068
##     TM4|t1         -4.584   0.874  -5.246   0.000  -4.584  -2.227
##     TM4|t2         -3.629   0.585  -6.201   0.000  -3.629  -1.763
##     TM4|t3         -2.921   0.470  -6.210   0.000  -2.921  -1.419
##     TM4|t4         -2.320   0.401  -5.781   0.000  -2.320  -1.127
##     TM4|t5         -1.496   0.325  -4.599   0.000  -1.496  -0.726
##     TM4|t6          0.303   0.298   1.017   0.309   0.303   0.147
##     TM4|t7          1.974   0.356   5.551   0.000   1.974   0.959
##     TM4|t8          3.118   0.455   6.857   0.000   3.118   1.515
##     TM5|t1         -4.416   0.887  -4.978   0.000  -4.416  -2.227
##     TM5|t2         -3.496   0.613  -5.700   0.000  -3.496  -1.763
##     TM5|t3         -2.648   0.413  -6.408   0.000  -2.648  -1.335
##     TM5|t4         -1.358   0.290  -4.685   0.000  -1.358  -0.685
##     TM5|t5         -0.357   0.277  -1.291   0.197  -0.357  -0.180
##     TM5|t6          0.556   0.296   1.878   0.060   0.556   0.280
##     TM5|t7          1.706   0.346   4.927   0.000   1.706   0.860
##     TM5|t8          3.225   0.460   7.007   0.000   3.225   1.626
##     CA2|t1         -4.028   0.557  -7.228   0.000  -4.028  -2.227
##     CA2|t2         -3.516   0.454  -7.743   0.000  -3.516  -1.944
##     CA2|t3         -3.189   0.395  -8.078   0.000  -3.189  -1.763
##     CA2|t4         -2.567   0.372  -6.906   0.000  -2.567  -1.419
##     CA2|t5         -1.734   0.332  -5.220   0.000  -1.734  -0.959
##     CA2|t6         -0.955   0.284  -3.363   0.001  -0.955  -0.528
##     CA2|t7          0.029   0.260   0.113   0.910   0.029   0.016
##     CA2|t8          1.239   0.278   4.460   0.000   1.239   0.685
##     CA2|t9          2.154   0.342   6.294   0.000   2.154   1.191
##     CA3|t1         -3.461   0.522  -6.630   0.000  -3.461  -2.227
##     CA3|t2         -3.021   0.400  -7.544   0.000  -3.021  -1.944
##     CA3|t3         -2.528   0.326  -7.750   0.000  -2.528  -1.626
##     CA3|t4         -1.412   0.263  -5.368   0.000  -1.412  -0.908
##     CA3|t5         -0.821   0.245  -3.354   0.001  -0.821  -0.528
##     CA3|t6          0.178   0.221   0.801   0.423   0.178   0.114
##     CA3|t7          1.412   0.260   5.423   0.000   1.412   0.908
##     CA3|t8          1.958   0.307   6.369   0.000   1.958   1.260
##     CA4|t1         -3.377   0.456  -7.407   0.000  -3.377  -1.944
##     CA4|t2         -2.826   0.411  -6.878   0.000  -2.826  -1.626
##     CA4|t3         -2.189   0.362  -6.046   0.000  -2.189  -1.260
##     CA4|t4         -1.190   0.304  -3.912   0.000  -1.190  -0.685
##     CA4|t5         -0.370   0.264  -1.404   0.160  -0.370  -0.213
##     CA4|t6          1.119   0.262   4.274   0.000   1.119   0.644
##     CA4|t7          2.826   0.420   6.724   0.000   2.826   1.626
##     CA5|t1         -2.931   0.424  -6.916   0.000  -2.931  -1.944
##     CA5|t2         -2.453   0.356  -6.895   0.000  -2.453  -1.626
##     CA5|t3         -1.446   0.267  -5.417   0.000  -1.446  -0.959
##     CA5|t4         -0.686   0.230  -2.980   0.003  -0.686  -0.455
##     CA5|t5          0.123   0.216   0.570   0.569   0.123   0.081
##     CA5|t6          0.972   0.215   4.519   0.000   0.972   0.644
##     CA5|t7          1.796   0.223   8.068   0.000   1.796   1.191
##     TH1|t1         -2.990   0.554  -5.399   0.000  -2.990  -2.227
##     TH1|t2         -2.184   0.339  -6.435   0.000  -2.184  -1.626
##     TH1|t3         -1.093   0.228  -4.792   0.000  -1.093  -0.814
##     TH1|t4         -0.153   0.195  -0.786   0.432  -0.153  -0.114
##     TH1|t5          0.865   0.193   4.492   0.000   0.865   0.644
##     TH1|t6          2.034   0.213   9.552   0.000   2.034   1.515
```

```
##     TH3|t1          -3.662    0.625   -5.861    0.000   -3.662   -2.227
##     TH3|t2          -3.196    0.482   -6.637    0.000   -3.196   -1.944
##     TH3|t3          -2.491    0.359   -6.937    0.000   -2.491   -1.515
##     TH3|t4          -1.059    0.267   -3.965    0.000   -1.059   -0.644
##     TH3|t5          -0.188    0.239   -0.786    0.432   -0.188   -0.114
##     TH3|t6           0.931    0.236    3.951    0.000    0.931    0.566
##     TH3|t7           1.854    0.234    7.933    0.000    1.854    1.127
##     TH4|t1          -2.698    0.449   -6.010    0.000   -2.698   -1.944
##     TH4|t2          -2.258    0.356   -6.336    0.000   -2.258   -1.626
##     TH4|t3          -1.749    0.292   -5.983    0.000   -1.749   -1.260
##     TH4|t4          -0.894    0.228   -3.922    0.000   -0.894   -0.644
##     TH4|t5          -0.159    0.202   -0.786    0.432   -0.159   -0.114
##     TH4|t6           0.839    0.208    4.029    0.000    0.839    0.605
##     TH4|t7           1.970    0.280    7.027    0.000    1.970    1.419
##     TH5|t1          -4.366    0.826   -5.284    0.000   -4.366   -2.227
##     TH5|t2          -3.457    0.511   -6.764    0.000   -3.457   -1.763
##     TH5|t3          -2.783    0.495   -5.617    0.000   -2.783   -1.419
##     TH5|t4          -2.093    0.434   -4.819    0.000   -2.093   -1.068
##     TH5|t5          -1.186    0.351   -3.375    0.001   -1.186   -0.605
##     TH5|t6          -0.032    0.283   -0.113    0.910   -0.032   -0.016
##     TH5|t7           1.110    0.324    3.430    0.001    1.110    0.566
##     TH5|t8           1.687    0.354    4.764    0.000    1.687    0.860
##     NE3|t1          -1.512    0.803   -1.883    0.060   -1.512   -0.825
##     NE4|t1          -0.647    0.270   -2.393    0.017   -0.647   -0.482
##     NE5|t1           0.071    0.181    0.394    0.693    0.071    0.066
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               1.000                               1.000    0.127
##    .AW2               1.000                               1.000    0.461
##    .AW3               1.000                               1.000    0.620
##    .AW4               1.000                               1.000    0.310
##    .TM1               1.000                               1.000    0.156
##    .TM2               1.000                               1.000    0.149
##    .TM3               1.000                               1.000    0.474
##    .TM4               1.000                               1.000    0.236
##    .TM5               1.000                               1.000    0.254
##    .CA2               1.000                               1.000    0.306
##    .CA3               1.000                               1.000    0.414
##    .CA4               1.000                               1.000    0.331
##    .CA5               1.000                               1.000    0.440
##    .TH1               1.000                               1.000    0.555
##    .TH3               1.000                               1.000    0.370
##    .TH4               1.000                               1.000    0.519
##    .TH5               1.000                               1.000    0.260
##    .NE3               1.000                               1.000    0.297
##    .NE4               1.000                               1.000    0.556
##    .NE5               1.000                               1.000    0.856
##     aw                6.862    3.553    1.931    0.053    1.000    1.000
##     tm                5.392    1.989    2.711    0.007    1.000    1.000
##     ca                2.272    0.655    3.471    0.001    1.000    1.000
##     th                0.803    0.259    3.098    0.002    1.000    1.000
##
## Scales y*:
```

```
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     AW1               0.357                               0.357    1.000
##     AW2               0.679                               0.679    1.000
##     AW3               0.788                               0.788    1.000
##     AW4               0.557                               0.557    1.000
##     TM1               0.396                               0.396    1.000
##     TM2               0.386                               0.386    1.000
##     TM3               0.689                               0.689    1.000
##     TM4               0.486                               0.486    1.000
##     TM5               0.504                               0.504    1.000
##     CA2               0.553                               0.553    1.000
##     CA3               0.643                               0.643    1.000
##     CA4               0.575                               0.575    1.000
##     CA5               0.663                               0.663    1.000
##     TH1               0.745                               0.745    1.000
##     TH3               0.608                               0.608    1.000
##     TH4               0.720                               0.720    1.000
##     TH5               0.510                               0.510    1.000
##     NE3               0.545                               0.545    1.000
##     NE4               0.745                               0.745    1.000
##     NE5               0.925                               0.925    1.000
```

Conclusion

Looking for guidance on how to appropriately model this data set given the number of missing observations and whether a reduced SEM, CFA, or maybe an alternative approach is most appropriate for our situation.

```
model4 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE5 ~ NE3 + NE4 + aw + tm + ca + th
"

fit4 <- sem(
  model4,
  data = SEMcombined,
  ordered = c("NE3","NE4","NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_data_full():
##    exogenous variable(s) declared as ordered in data: "NE3", "NE4"

## Warning: lavaan->lav_data_full():
##    30 cases were deleted due to missing values in exogenous variable(s),
##    while fixed.x = TRUE.

## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 47 48.
```

```
## Warning: lavaan->lav_model_vcov():
##    The variance-covariance matrix of the estimated parameters (vcov) does not
##    appear to be positive definite! The smallest eigenvalue (= -7.980236e-15)
##    is smaller than zero. This may be a symptom that the model is not
##    identified.
```

```r
summary(fit4, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 70 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        64
##
##                                                   Used       Total
##   Number of observations                            52          84
##   Number of missing patterns                         3
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               100.971     187.754
##   Degrees of freedom                               160         160
##   P-value (Unknown)                                 NA       0.066
##   Scaling correction factor                                  1.328
##   Shift parameter                                          111.743
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              1311.977     389.295
##   Degrees of freedom                               153         153
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.905
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.883
##   Tucker-Lewis Index (TLI)                       1.049       0.888
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.058
##   90 Percent confidence interval - lower         0.000       0.000
##   90 Percent confidence interval - upper         0.000       0.090
##   P-value H_0: RMSEA <= 0.050                    1.000       0.345
##   P-value H_0: RMSEA >= 0.080                    0.000       0.142
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
```

```
##   P-value H_0: Robust RMSEA >= 0.080                             NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.100       0.100
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.690    0.945
##     AW2               0.809    0.176    4.599    0.000    1.367    0.716
##     AW3               0.449    0.113    3.971    0.000    0.759    0.494
##     AW4               0.956    0.192    4.981    0.000    1.616    0.796
##   tm =~
##     TM1               1.000                               2.015    0.976
##     TM2               0.892    0.099    9.024    0.000    1.798    0.918
##     TM3               0.642    0.116    5.548    0.000    1.294    0.666
##     TM4               0.766    0.095    8.032    0.000    1.543    0.851
##     TM5               0.898    0.127    7.082    0.000    1.810    0.915
##   ca =~
##     CA2               1.000                               1.422    0.726
##     CA3               0.990    0.170    5.807    0.000    1.407    0.728
##     CA4               0.891    0.233    3.832    0.000    1.267    0.814
##     CA5               0.844    0.191    4.421    0.000    1.200    0.716
##   th =~
##     TH1               1.000                               0.745    0.548
##     TH3               1.553    0.282    5.497    0.000    1.156    0.727
##     TH4               1.552    0.466    3.327    0.001    1.155    0.713
##     TH5               2.116    0.445    4.750    0.000    1.575    0.843
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE5 ~
##     NE3              -0.214    0.542   -0.395    0.693   -0.214   -0.083
##     NE4              -0.227    0.449   -0.505    0.613   -0.227   -0.101
##     aw                0.020    0.287    0.069    0.945    0.034    0.032
##     tm                0.088    0.149    0.589    0.556    0.177    0.169
##     ca                0.112    0.282    0.399    0.690    0.160    0.152
##     th               -0.072    0.707   -0.102    0.919   -0.054   -0.051
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     tm                1.779    0.565    3.150    0.002    0.522    0.522
##     ca                1.613    0.557    2.895    0.004    0.671    0.671
##     th                1.099    0.302    3.634    0.000    0.873    0.873
##   tm ~~
```

```
##     ca                2.095    0.584    3.591    0.000    0.731    0.731
##     th                0.897    0.298    3.010    0.003    0.598    0.598
##   ca ~~
##     th                0.800    0.260    3.080    0.002    0.755    0.755
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               8.842    1.992    4.440    0.000    8.842    4.946
##    .AW2               8.552    1.870    4.574    0.000    8.552    4.482
##    .AW3              10.032    1.814    5.530    0.000   10.032    6.532
##    .AW4               8.121    1.997    4.066    0.000    8.121    4.001
##    .TM1               6.696    2.577    2.598    0.009    6.696    3.245
##    .TM2               8.813    2.484    3.548    0.000    8.813    4.501
##    .TM3               7.668    2.286    3.354    0.001    7.668    3.950
##    .TM4               6.623    2.915    2.272    0.023    6.623    3.652
##    .TM5               6.972    2.122    3.286    0.001    6.972    3.526
##    .CA2               8.894    2.032    4.376    0.000    8.894    4.542
##    .CA3               8.385    2.056    4.079    0.000    8.385    4.339
##    .CA4              10.128    2.642    3.833    0.000   10.128    6.511
##    .CA5              11.214    2.291    4.895    0.000   11.214    6.694
##    .TH1               9.332    1.434    6.507    0.000    9.332    6.865
##    .TH3               9.838    1.838    5.351    0.000    9.838    6.185
##    .TH4               8.998    1.635    5.505    0.000    8.998    5.552
##    .TH5               8.311    1.981    4.195    0.000    8.311    4.447
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5|t1           -0.640    1.387   -0.462    0.644   -0.640   -0.610
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.339    0.316    1.073    0.283    0.339    0.106
##    .AW2               1.772    0.431    4.114    0.000    1.772    0.487
##    .AW3               1.782    0.318    5.612    0.000    1.782    0.756
##    .AW4               1.508    0.448    3.369    0.001    1.508    0.366
##    .TM1               0.198    0.392    0.505    0.613    0.198    0.047
##    .TM2               0.602    0.268    2.248    0.025    0.602    0.157
##    .TM3               2.095    0.341    6.137    0.000    2.095    0.556
##    .TM4               0.910    0.337    2.701    0.007    0.910    0.277
##    .TM5               0.634    0.367    1.727    0.084    0.634    0.162
##    .CA2               1.812    0.288    6.299    0.000    1.812    0.473
##    .CA3               1.753    0.309    5.667    0.000    1.753    0.470
##    .CA4               0.815    0.216    3.780    0.000    0.815    0.337
##    .CA5               1.366    0.286    4.777    0.000    1.366    0.487
##    .TH1               1.294    0.214    6.054    0.000    1.294    0.700
##    .TH3               1.193    0.179    6.671    0.000    1.193    0.472
##    .TH4               1.292    0.305    4.238    0.000    1.292    0.492
##    .TH5               1.011    0.394    2.567    0.010    1.011    0.289
##    .NE5               1.000                               1.000    0.907
##     aw                2.857    0.896    3.186    0.001    1.000    1.000
##     tm                4.060    1.194    3.401    0.001    1.000    1.000
##     ca                2.022    0.802    2.520    0.012    1.000    1.000
##     th                0.554    0.221    2.507    0.012    1.000    1.000
##
```

```
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5               0.959                               0.959    1.000
```

```r
model5 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NI =~ aw + tm + ca + th
  NE3 ~~ NE4
  NE3 ~~ NI
  NE4 ~~ NI
  NE5 ~ NE3 + NE4 + NI
"

fit5 <- sem(
  model5,
  data = SEMcombined,
  ordered = c("NE3","NE4","NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)

summary(fit5, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 58 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        64
##
##   Number of observations                            84
##   Number of missing patterns                         8
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               149.172     271.384
##   Degrees of freedom                               163         163
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.894
##   Shift parameter                                          104.595
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              2355.400     663.098
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.577
##
## User Model versus Baseline Model:
```

```
##
##   Comparative Fit Index (CFI)                    1.000       0.771
##   Tucker-Lewis Index (TLI)                       1.007       0.733
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.090
##   90 Percent confidence interval - lower         0.000       0.070
##   90 Percent confidence interval - upper         0.036       0.108
##   P-value H_0: RMSEA <= 0.050                    0.992       0.001
##   P-value H_0: RMSEA >= 0.080                    0.000       0.803
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.112       0.112
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                          Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.697    0.960
##     AW2               0.795    0.138    5.762    0.000    1.350    0.696
##     AW3               0.544    0.096    5.690    0.000    0.923    0.570
##     AW4               0.902    0.134    6.717    0.000    1.531    0.796
##   tm =~
##     TM1               1.000                               1.843    0.936
##     TM2               1.049    0.097   10.831    0.000    1.933    0.917
##     TM3               0.772    0.078    9.907    0.000    1.423    0.701
##     TM4               0.795    0.077   10.324    0.000    1.465    0.851
##     TM5               0.922    0.093    9.965    0.000    1.698    0.906
##   ca =~
##     CA2               1.000                               1.491    0.785
##     CA3               0.829    0.097    8.542    0.000    1.236    0.677
##     CA4               0.787    0.131    6.027    0.000    1.173    0.756
##     CA5               0.808    0.131    6.154    0.000    1.205    0.692
##   th =~
##     TH1               1.000                               0.905    0.646
##     TH3               1.324    0.209    6.320    0.000    1.198    0.757
##     TH4               1.222    0.251    4.868    0.000    1.106    0.691
```

```
##     TH5               1.799    0.324    5.555    0.000    1.628    0.855
##   NI =~
##     aw                1.000                               0.848    0.848
##     tm                0.911    0.164    5.542    0.000    0.712    0.712
##     ca                1.000    0.171    5.843    0.000    0.965    0.965
##     th                0.470    0.079    5.943    0.000    0.748    0.748
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE5 ~
##     NE3               0.181    0.366    0.494    0.621    0.181    0.171
##     NE4               0.025    0.388    0.063    0.949    0.025    0.023
##     NI                0.241    0.154    1.564    0.118    0.346    0.326
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~~
##     NE4              -0.509    0.227   -2.249    0.025   -0.509   -0.509
##   NI ~~
##     NE3              -0.241    0.202   -1.190    0.234   -0.167   -0.167
##     NE4              -0.186    0.239   -0.778    0.436   -0.129   -0.129
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.312    0.234   31.297    0.000    7.312    4.137
##    .AW2               7.221    0.240   30.059    0.000    7.221    3.726
##    .AW3               7.948    0.211   37.685    0.000    7.948    4.910
##    .AW4               7.065    0.238   29.685    0.000    7.065    3.674
##    .TM1               7.091    0.247   28.713    0.000    7.091    3.602
##    .TM2               6.909    0.284   24.302    0.000    6.909    3.277
##    .TM3               7.506    0.296   25.360    0.000    7.506    3.698
##    .TM4               7.156    0.236   30.377    0.000    7.156    4.157
##    .TM5               6.792    0.231   29.452    0.000    6.792    3.623
##    .CA2               7.234    0.246   29.401    0.000    7.234    3.808
##    .CA3               7.130    0.235   30.381    0.000    7.130    3.905
##    .CA4               7.468    0.207   36.024    0.000    7.468    4.815
##    .CA5               7.273    0.205   35.521    0.000    7.273    4.178
##    .TH1               7.571    0.174   43.578    0.000    7.571    5.409
##    .TH3               7.584    0.191   39.630    0.000    7.584    4.794
##    .TH4               7.455    0.203   36.769    0.000    7.455    4.659
##    .TH5               7.442    0.236   31.537    0.000    7.442    3.909
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5|t1            0.070    0.177    0.395    0.693    0.070    0.066
##     NE3|t1           -0.825    0.157   -5.254    0.000   -0.825   -0.825
##     NE4|t1           -0.482    0.179   -2.694    0.007   -0.482   -0.482
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.242    0.241    1.006    0.314    0.242    0.078
##    .AW2               1.935    0.375    5.158    0.000    1.935    0.515
##    .AW3               1.769    0.260    6.809    0.000    1.769    0.675
##    .AW4               1.354    0.335    4.038    0.000    1.354    0.366
```

```
##    .TM1               0.480    0.250    1.922    0.055    0.480    0.124
##    .TM2               0.708    0.211    3.352    0.001    0.708    0.159
##    .TM3               2.094    0.214    9.767    0.000    2.094    0.508
##    .TM4               0.816    0.176    4.631    0.000    0.816    0.276
##    .TM5               0.631    0.233    2.704    0.007    0.631    0.179
##    .CA2               1.385    0.179    7.731    0.000    1.385    0.384
##    .CA3               1.805    0.173   10.446    0.000    1.805    0.542
##    .CA4               1.029    0.146    7.053    0.000    1.029    0.428
##    .CA5               1.577    0.241    6.544    0.000    1.577    0.520
##    .TH1               1.140    0.137    8.319    0.000    1.140    0.582
##    .TH3               1.068    0.142    7.497    0.000    1.068    0.427
##    .TH4               1.336    0.251    5.322    0.000    1.336    0.522
##    .TH5               0.972    0.330    2.949    0.003    0.972    0.268
##    .NE5               1.000                               1.000    0.888
##     NE3               1.000                               1.000    1.000
##     NE4               1.000                               1.000    1.000
##    .aw                0.809    0.214    3.778    0.000    0.281    0.281
##    .tm                1.675    0.348    4.820    0.000    0.493    0.493
##    .ca                0.151    0.172    0.876    0.381    0.068    0.068
##    .th                0.361    0.145    2.484    0.013    0.440    0.440
##     NI                2.072    0.543    3.814    0.000    1.000    1.000
##
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5               0.943                               0.943    1.000
```

```r
model6 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NI =~ aw + tm + ca + th
  NE4 ~~ NI
  NE5 ~ NE4 + NI
"

fit6 <- sem(
  model6,
  data = SEMcombined,
  ordered = c("NE4","NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 21 30 70.
```

```r
summary(fit6, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 55 iterations
##
```

```
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        60
##
##                                                   Used       Total
##   Number of observations                            81          84
##   Number of missing patterns                         6
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               118.490     252.407
##   Degrees of freedom                               147         147
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.732
##   Shift parameter                                           90.550
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              2310.329     627.398
##   Degrees of freedom                               171         171
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.687
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.769
##   Tucker-Lewis Index (TLI)                       1.016       0.731
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.095
##   90 Percent confidence interval - lower         0.000       0.075
##   90 Percent confidence interval - upper         0.000       0.114
##   P-value H_0: RMSEA <= 0.050                    0.999       0.000
##   P-value H_0: RMSEA >= 0.080                    0.000       0.889
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.097       0.097
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                           Robust.sem
```

```
##   Information                                 Expected
##   Information saturated (h1) model          Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.696    0.960
##     AW2               0.792    0.138    5.729    0.000    1.343    0.693
##     AW3               0.546    0.096    5.711    0.000    0.926    0.572
##     AW4               0.906    0.134    6.769    0.000    1.536    0.799
##   tm =~
##     TM1               1.000                               1.845    0.937
##     TM2               1.046    0.097   10.825    0.000    1.930    0.915
##     TM3               0.769    0.078    9.832    0.000    1.419    0.699
##     TM4               0.794    0.078   10.241    0.000    1.464    0.851
##     TM5               0.924    0.093    9.975    0.000    1.704    0.909
##   ca =~
##     CA2               1.000                               1.490    0.784
##     CA3               0.834    0.096    8.681    0.000    1.243    0.681
##     CA4               0.786    0.130    6.044    0.000    1.170    0.755
##     CA5               0.806    0.132    6.109    0.000    1.201    0.690
##   th =~
##     TH1               1.000                               0.905    0.646
##     TH3               1.324    0.211    6.289    0.000    1.198    0.757
##     TH4               1.219    0.251    4.860    0.000    1.103    0.689
##     TH5               1.804    0.326    5.535    0.000    1.632    0.857
##   NI =~
##     aw                1.000                               0.848    0.848
##     tm                0.920    0.165    5.588    0.000    0.717    0.717
##     ca                0.999    0.169    5.904    0.000    0.965    0.965
##     th                0.466    0.079    5.915    0.000    0.741    0.741
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE5 ~
##     NE4              -0.072    0.258   -0.279    0.780   -0.072   -0.069
##     NI                0.209    0.143    1.462    0.144    0.301    0.287
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NI ~~
##     NE4              -0.187    0.239   -0.785    0.432   -0.130   -0.130
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.312    0.234   31.290    0.000    7.312    4.137
##    .AW2               7.221    0.240   30.052    0.000    7.221    3.726
##    .AW3               7.948    0.211   37.676    0.000    7.948    4.910
##    .AW4               7.065    0.238   29.679    0.000    7.065    3.674
##    .TM1               7.091    0.247   28.707    0.000    7.091    3.602
##    .TM2               6.909    0.284   24.297    0.000    6.909    3.277
##    .TM3               7.506    0.296   25.354    0.000    7.506    3.698
##    .TM4               7.156    0.236   30.371    0.000    7.156    4.157
##    .TM5               6.792    0.231   29.446    0.000    6.792    3.623
```

```
##    .CA2               7.234    0.246   29.395    0.000    7.234    3.808
##    .CA3               7.130    0.235   30.374    0.000    7.130    3.905
##    .CA4               7.468    0.207   36.016    0.000    7.468    4.815
##    .CA5               7.273    0.205   35.513    0.000    7.273    4.178
##    .TH1               7.571    0.174   43.568    0.000    7.571    5.409
##    .TH3               7.584    0.191   39.622    0.000    7.584    4.794
##    .TH4               7.455    0.203   36.760    0.000    7.455    4.659
##    .TH5               7.442    0.236   31.530    0.000    7.442    3.909
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5|t1            0.069    0.176    0.395    0.693    0.069    0.066
##     NE4|t1           -0.482    0.179   -2.693    0.007   -0.482   -0.482
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.248    0.240    1.034    0.301    0.248    0.079
##    .AW2               1.953    0.377    5.184    0.000    1.953    0.520
##    .AW3               1.764    0.259    6.802    0.000    1.764    0.673
##    .AW4               1.337    0.334    4.001    0.000    1.337    0.362
##    .TM1               0.472    0.252    1.874    0.061    0.472    0.122
##    .TM2               0.723    0.208    3.482    0.000    0.723    0.163
##    .TM3               2.105    0.214    9.827    0.000    2.105    0.511
##    .TM4               0.818    0.176    4.656    0.000    0.818    0.276
##    .TM5               0.613    0.230    2.665    0.008    0.613    0.174
##    .CA2               1.388    0.181    7.658    0.000    1.388    0.385
##    .CA3               1.788    0.173   10.360    0.000    1.788    0.536
##    .CA4               1.035    0.147    7.036    0.000    1.035    0.430
##    .CA5               1.588    0.245    6.487    0.000    1.588    0.524
##    .TH1               1.141    0.138    8.262    0.000    1.141    0.582
##    .TH3               1.068    0.143    7.458    0.000    1.068    0.427
##    .TH4               1.343    0.252    5.324    0.000    1.343    0.525
##    .TH5               0.961    0.333    2.885    0.004    0.961    0.265
##    .NE5               1.000                               1.000    0.908
##     NE4               1.000                               1.000    1.000
##    .aw                0.807    0.212    3.801    0.000    0.281    0.281
##    .tm                1.653    0.343    4.822    0.000    0.486    0.486
##    .ca                0.154    0.175    0.880    0.379    0.069    0.069
##    .th                0.369    0.147    2.503    0.012    0.451    0.451
##     NI                2.069    0.540    3.831    0.000    1.000    1.000
##
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5               0.953                               0.953    1.000
```

```
model7 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE4 ~~ aw + tm + ca + th
  NE5 ~ NE4 + aw + tm + ca + th
"
```

```r
fit7 <- sem(
  model7,
  data = SEMcombined,
  ordered = c("NE4","NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_data_full():
##    some cases are empty and will be ignored: 21 30 70.
```

```r
summary(fit7, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 65 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        68
##
##                                                   Used       Total
##   Number of observations                            81          84
##   Number of missing patterns                         6
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                                90.323     240.310
##   Degrees of freedom                               139         139
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.571
##   Shift parameter                                           82.175
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              2310.329     627.398
##   Degrees of freedom                               171         171
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.687
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.778
##   Tucker-Lewis Index (TLI)                       1.028       0.727
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.095
##   90 Percent confidence interval - lower         0.000       0.075
```

```
##   90 Percent confidence interval - upper         0.000       0.115
##   P-value H_0: RMSEA <= 0.050                    1.000       0.000
##   P-value H_0: RMSEA >= 0.080                    0.000       0.895
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.084       0.084
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.693    0.958
##     AW2               0.811    0.139    5.814    0.000    1.372    0.708
##     AW3               0.537    0.093    5.766    0.000    0.909    0.561
##     AW4               0.902    0.131    6.858    0.000    1.526    0.794
##   tm =~
##     TM1               1.000                               1.850    0.940
##     TM2               1.051    0.098   10.668    0.000    1.943    0.922
##     TM3               0.770    0.079    9.795    0.000    1.424    0.702
##     TM4               0.786    0.077   10.257    0.000    1.453    0.844
##     TM5               0.917    0.092    9.989    0.000    1.696    0.905
##   ca =~
##     CA2               1.000                               1.460    0.769
##     CA3               0.841    0.097    8.637    0.000    1.228    0.673
##     CA4               0.809    0.133    6.066    0.000    1.182    0.762
##     CA5               0.842    0.136    6.190    0.000    1.230    0.707
##   th =~
##     TH1               1.000                               0.895    0.639
##     TH3               1.358    0.211    6.437    0.000    1.216    0.768
##     TH4               1.226    0.250    4.894    0.000    1.097    0.686
##     TH5               1.825    0.322    5.669    0.000    1.633    0.858
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE5 ~
##     NE4              -0.075    0.353   -0.211    0.833   -0.075   -0.070
##     aw               -0.102    0.292   -0.348    0.728   -0.172   -0.160
##     tm                0.103    0.200    0.516    0.606    0.191    0.178
##     ca                0.223    0.345    0.646    0.518    0.326    0.304
##     th               -0.029    0.561   -0.051    0.959   -0.026   -0.024
##
```

```
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     NE4              -0.335    0.286   -1.169    0.242   -0.198   -0.198
##   tm ~~
##     NE4              -0.306    0.232   -1.315    0.188   -0.165   -0.165
##   ca ~~
##     NE4              -0.171    0.312   -0.547    0.584   -0.117   -0.117
##   th ~~
##     NE4               0.119    0.138    0.863    0.388    0.133    0.133
##   aw ~~
##     tm                1.865    0.419    4.451    0.000    0.596    0.596
##     ca                1.789    0.433    4.128    0.000    0.724    0.724
##     th                1.174    0.252    4.652    0.000    0.775    0.775
##   tm ~~
##     ca                2.051    0.421    4.874    0.000    0.759    0.759
##     th                0.775    0.216    3.584    0.000    0.468    0.468
##   ca ~~
##     th                0.923    0.215    4.299    0.000    0.706    0.706
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.312    0.234   31.290    0.000    7.312    4.137
##    .AW2               7.221    0.240   30.052    0.000    7.221    3.726
##    .AW3               7.948    0.211   37.676    0.000    7.948    4.910
##    .AW4               7.065    0.238   29.679    0.000    7.065    3.674
##    .TM1               7.091    0.247   28.707    0.000    7.091    3.602
##    .TM2               6.909    0.284   24.297    0.000    6.909    3.277
##    .TM3               7.506    0.296   25.354    0.000    7.506    3.698
##    .TM4               7.156    0.236   30.371    0.000    7.156    4.157
##    .TM5               6.792    0.231   29.446    0.000    6.792    3.623
##    .CA2               7.234    0.246   29.395    0.000    7.234    3.808
##    .CA3               7.130    0.235   30.374    0.000    7.130    3.905
##    .CA4               7.468    0.207   36.016    0.000    7.468    4.815
##    .CA5               7.273    0.205   35.513    0.000    7.273    4.178
##    .TH1               7.571    0.174   43.568    0.000    7.571    5.409
##    .TH3               7.584    0.191   39.622    0.000    7.584    4.794
##    .TH4               7.455    0.203   36.760    0.000    7.455    4.659
##    .TH5               7.442    0.236   31.530    0.000    7.442    3.909
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5|t1            0.071    0.179    0.395    0.693    0.071    0.066
##     NE4|t1           -0.482    0.179   -2.693    0.007   -0.482   -0.482
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.258    0.230    1.120    0.263    0.258    0.083
##    .AW2               1.874    0.361    5.186    0.000    1.874    0.499
##    .AW3               1.795    0.258    6.964    0.000    1.795    0.685
##    .AW4               1.368    0.325    4.212    0.000    1.368    0.370
##    .TM1               0.454    0.254    1.790    0.074    0.454    0.117
##    .TM2               0.670    0.203    3.300    0.001    0.670    0.151
##    .TM3               2.092    0.208   10.038    0.000    2.092    0.508
```

```
##    .TM4               0.852    0.172    4.951    0.000    0.852    0.288
##    .TM5               0.638    0.227    2.813    0.005    0.638    0.181
##    .CA2               1.475    0.177    8.324    0.000    1.475    0.409
##    .CA3               1.825    0.168   10.847    0.000    1.825    0.548
##    .CA4               1.008    0.144    6.991    0.000    1.008    0.419
##    .CA5               1.517    0.235    6.444    0.000    1.517    0.501
##    .TH1               1.158    0.136    8.539    0.000    1.158    0.591
##    .TH3               1.025    0.140    7.317    0.000    1.025    0.410
##    .TH4               1.357    0.250    5.426    0.000    1.357    0.530
##    .TH5               0.955    0.317    3.010    0.003    0.955    0.264
##    .NE5               1.000                               1.000    0.870
##     NE4               1.000                               1.000    1.000
##     aw                2.865    0.638    4.491    0.000    1.000    1.000
##     tm                3.421    0.766    4.465    0.000    1.000    1.000
##     ca                2.133    0.546    3.908    0.000    1.000    1.000
##     th                0.801    0.238    3.368    0.001    1.000    1.000
##
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE5               0.933                               0.933    1.000
```

```r
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode

## The following object is masked from 'package:purrr':
##
##     some
```

```r
SEMcombined$aw <- rowMeans(SEMcombined[, c("AW1","AW2","AW3","AW4")], na.rm = TRUE)
SEMcombined$tm <- rowMeans(SEMcombined[, c("TM1","TM2","TM3","TM4","TM5")], na.rm = TRUE)
SEMcombined$ca <- rowMeans(SEMcombined[, c("CA2","CA3","CA4","CA5")], na.rm = TRUE)
SEMcombined$th <- rowMeans(SEMcombined[, c("TH1","TH3","TH4","TH5")], na.rm = TRUE)

A3_log <- glm(NE3 ~ aw + tm + ca + th, data = SEMcombined, family = binomial)
A4_log <- glm(NE4 ~ aw + tm + ca + th, data = SEMcombined, family = binomial)

model8 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE3 ~ aw + tm + ca + th
  NE4 ~ NE3 + aw + tm + ca + th
```

```r
  NE5 ~ NE4 + aw + tm + ca + th
"

fit8 <- sem(
  model8,
  data = SEMcombined,
  ordered = c("NE3", "NE4", "NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_lavaan_step01_ovnames_checklv():
##    Some latent variable names collide with observed variable names in the
##    dataset: aw tm ca th. Please provide alternative names for the latent
##    variables, or switch off this check using check.lv.names = FALSE
```

```r
summary(fit8, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 100 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        74
##
##   Number of observations                            84
##   Number of missing patterns                         8
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               102.227     247.118
##   Degrees of freedom                               153         153
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.666
##   Shift parameter                                           93.535
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              2355.400     663.098
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.577
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.801
##   Tucker-Lewis Index (TLI)                       1.029       0.753
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
```

```
##
##   RMSEA                                          0.000       0.086
##   90 Percent confidence interval - lower         0.000       0.066
##   90 Percent confidence interval - upper         0.000       0.105
##   P-value H_0: RMSEA <= 0.050                    1.000       0.003
##   P-value H_0: RMSEA >= 0.080                    0.000       0.704
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.089       0.089
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.695    0.959
##     AW2               0.812    0.139    5.833    0.000    1.377    0.710
##     AW3               0.536    0.093    5.749    0.000    0.908    0.561
##     AW4               0.898    0.132    6.813    0.000    1.522    0.791
##   tm =~
##     TM1               1.000                               1.848    0.939
##     TM2               1.050    0.098   10.684    0.000    1.940    0.920
##     TM3               0.769    0.079    9.752    0.000    1.422    0.701
##     TM4               0.787    0.077   10.197    0.000    1.455    0.845
##     TM5               0.919    0.092   10.022    0.000    1.699    0.906
##   ca =~
##     CA2               1.000                               1.459    0.768
##     CA3               0.836    0.099    8.457    0.000    1.220    0.668
##     CA4               0.814    0.135    6.042    0.000    1.188    0.766
##     CA5               0.848    0.136    6.241    0.000    1.237    0.711
##   th =~
##     TH1               1.000                               0.905    0.647
##     TH3               1.349    0.207    6.519    0.000    1.221    0.772
##     TH4               1.223    0.249    4.914    0.000    1.107    0.692
##     TH5               1.773    0.311    5.700    0.000    1.604    0.843
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~
##     aw                0.473    0.272    1.740    0.082    0.802    0.566
##     tm                0.334    0.323    1.034    0.301    0.618    0.436
##     ca               -0.292    0.487   -0.599    0.549   -0.425   -0.300
```

```
##     th                -1.432    0.853   -1.677    0.093   -1.296   -0.915
##   NE4 ~
##     NE3               -0.440    0.429   -1.026    0.305   -0.440   -0.493
##     aw                -0.291    0.301   -0.967    0.333   -0.494   -0.390
##     tm                 0.161    0.258    0.624    0.532    0.298    0.236
##     ca                -0.301    0.484   -0.622    0.534   -0.439   -0.347
##     th                 0.460    0.645    0.713    0.476    0.416    0.329
##   NE5 ~
##     NE4               -0.076    0.275   -0.275    0.783   -0.076   -0.089
##     aw                -0.089    0.290   -0.307    0.759   -0.151   -0.140
##     tm                 0.112    0.196    0.574    0.566    0.208    0.194
##     ca                 0.201    0.335    0.601    0.548    0.293    0.274
##     th                -0.039    0.552   -0.070    0.944   -0.035   -0.033
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw ~~
##     tm                1.867    0.420    4.442    0.000    0.596    0.596
##     ca                1.783    0.432    4.127    0.000    0.721    0.721
##     th                1.186    0.253    4.691    0.000    0.773    0.773
##   tm ~~
##     ca                2.047    0.422    4.853    0.000    0.759    0.759
##     th                0.783    0.218    3.599    0.000    0.468    0.468
##   ca ~~
##     th                0.932    0.216    4.317    0.000    0.706    0.706
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.312    0.234   31.297    0.000    7.312    4.137
##    .AW2               7.221    0.240   30.059    0.000    7.221    3.726
##    .AW3               7.948    0.211   37.685    0.000    7.948    4.910
##    .AW4               7.065    0.238   29.685    0.000    7.065    3.674
##    .TM1               7.091    0.247   28.713    0.000    7.091    3.602
##    .TM2               6.909    0.284   24.302    0.000    6.909    3.277
##    .TM3               7.506    0.296   25.360    0.000    7.506    3.698
##    .TM4               7.156    0.236   30.377    0.000    7.156    4.157
##    .TM5               6.792    0.231   29.452    0.000    6.792    3.623
##    .CA2               7.234    0.246   29.401    0.000    7.234    3.808
##    .CA3               7.130    0.235   30.381    0.000    7.130    3.905
##    .CA4               7.468    0.207   36.024    0.000    7.468    4.815
##    .CA5               7.273    0.205   35.521    0.000    7.273    4.178
##    .TH1               7.571    0.174   43.578    0.000    7.571    5.409
##    .TH3               7.584    0.191   39.630    0.000    7.584    4.794
##    .TH4               7.455    0.203   36.769    0.000    7.455    4.659
##    .TH5               7.442    0.236   31.537    0.000    7.442    3.909
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3|t1           -1.168    0.310   -3.763    0.000   -1.168   -0.825
##     NE4|t1           -0.610    0.246   -2.482    0.013   -0.610   -0.482
##     NE5|t1            0.071    0.179    0.395    0.693    0.071    0.066
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##    .AW1               0.251    0.233    1.079    0.280    0.251    0.080
##    .AW2               1.861    0.361    5.156    0.000    1.861    0.495
##    .AW3               1.796    0.258    6.969    0.000    1.796    0.685
##    .AW4               1.382    0.325    4.249    0.000    1.382    0.374
##    .TM1               0.458    0.254    1.808    0.071    0.458    0.118
##    .TM2               0.683    0.201    3.404    0.001    0.683    0.154
##    .TM3               2.098    0.209   10.053    0.000    2.098    0.509
##    .TM4               0.846    0.172    4.926    0.000    0.846    0.286
##    .TM5               0.630    0.223    2.820    0.005    0.630    0.179
##    .CA2               1.480    0.175    8.447    0.000    1.480    0.410
##    .CA3               1.846    0.168   10.964    0.000    1.846    0.554
##    .CA4               0.994    0.143    6.967    0.000    0.994    0.414
##    .CA5               1.499    0.230    6.505    0.000    1.499    0.495
##    .TH1               1.140    0.134    8.516    0.000    1.140    0.582
##    .TH3               1.011    0.140    7.242    0.000    1.011    0.404
##    .TH4               1.334    0.246    5.414    0.000    1.334    0.521
##    .TH5               1.049    0.302    3.476    0.001    1.049    0.289
##    .NE3               1.000                               1.000    0.499
##    .NE4               1.000                               1.000    0.625
##    .NE5               1.000                               1.000    0.870
##     aw                2.872    0.640    4.486    0.000    1.000    1.000
##     tm                3.416    0.766    4.462    0.000    1.000    1.000
##     ca                2.127    0.547    3.892    0.000    1.000    1.000
##     th                0.819    0.240    3.414    0.001    1.000    1.000
##
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3               0.706                               0.706    1.000
##     NE4               0.790                               0.790    1.000
##     NE5               0.933                               0.933    1.000
```

```
model9 <- "
  aw =~ AW1 + AW2 + AW3 + AW4
  tm =~ TM1 + TM2 + TM3 + TM4 + TM5
  ca =~ CA2 + CA3 + CA4 + CA5
  th =~ TH1 + TH3 + TH4 + TH5

  NE3 ~ aw + tm + ca + th
  NE4 ~ aw + tm + ca + th
  NE5 ~ NE3 + NE4 + aw + tm + ca + th
"

fit9 <- sem(
  model9,
  data = SEMcombined,
  ordered = c("NE3", "NE4", "NE5"),
  estimator = "WLSMV",
  parameterization = "theta",
  missing = "pairwise"
)
```

```
## Warning: lavaan->lav_lavaan_step01_ovnames_checklv():
##    Some latent variable names collide with observed variable names in the
##    dataset: aw tm ca th. Please provide alternative names for the latent
```

```
##    variables, or switch off this check using check.lv.names = FALSE
```

```r
summary(fit9, fit.measures = TRUE, standardized = TRUE)
```

```
## lavaan 0.6-21 ended normally after 114 iterations
##
##   Estimator                                       DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        74
##
##   Number of observations                            84
##   Number of missing patterns                         8
##
## Model Test User Model:
##                                               Standard      Scaled
##   Test Statistic                               102.716     248.347
##   Degrees of freedom                               153         153
##   P-value (Unknown)                                 NA       0.000
##   Scaling correction factor                                  0.664
##   Shift parameter                                           93.717
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                              2355.400     663.098
##   Degrees of freedom                               190         190
##   P-value                                           NA       0.000
##   Scaling correction factor                                  4.577
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    1.000       0.798
##   Tucker-Lewis Index (TLI)                       1.029       0.750
##
##   Robust Comparative Fit Index (CFI)                            NA
##   Robust Tucker-Lewis Index (TLI)                               NA
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.000       0.087
##   90 Percent confidence interval - lower         0.000       0.066
##   90 Percent confidence interval - upper         0.000       0.106
##   P-value H_0: RMSEA <= 0.050                    1.000       0.003
##   P-value H_0: RMSEA >= 0.080                    0.000       0.720
##
##   Robust RMSEA                                                  NA
##   90 Percent confidence interval - lower                        NA
##   90 Percent confidence interval - upper                        NA
##   P-value H_0: Robust RMSEA <= 0.050                            NA
##   P-value H_0: Robust RMSEA >= 0.080                            NA
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.089       0.089
```

```
##
## Parameter Estimates:
##
##   Parameterization                               Theta
##   Standard errors                           Robust.sem
##   Information                                 Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   aw =~
##     AW1               1.000                               1.680    0.951
##     AW2               0.814    0.140    5.837    0.000    1.368    0.706
##     AW3               0.536    0.093    5.762    0.000    0.900    0.556
##     AW4               0.899    0.131    6.840    0.000    1.511    0.786
##   tm =~
##     TM1               1.000                               1.849    0.939
##     TM2               1.050    0.098   10.681    0.000    1.940    0.920
##     TM3               0.769    0.079    9.751    0.000    1.422    0.701
##     TM4               0.787    0.077   10.196    0.000    1.455    0.845
##     TM5               0.919    0.092   10.020    0.000    1.699    0.906
##   ca =~
##     CA2               1.000                               1.459    0.768
##     CA3               0.836    0.099    8.464    0.000    1.220    0.668
##     CA4               0.814    0.135    6.043    0.000    1.187    0.765
##     CA5               0.848    0.136    6.242    0.000    1.237    0.711
##   th =~
##     TH1               1.000                               0.894    0.639
##     TH3               1.348    0.207    6.524    0.000    1.205    0.762
##     TH4               1.224    0.249    4.919    0.000    1.094    0.684
##     TH5               1.774    0.311    5.714    0.000    1.586    0.833
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##   NE3 ~
##     aw                0.781    0.475    1.642    0.101    1.312    0.802
##     tm                0.319    0.371    0.862    0.389    0.591    0.361
##     ca               -0.258    0.557   -0.463    0.643   -0.376   -0.230
##     th               -2.131    1.300   -1.638    0.101   -1.905   -1.165
##   NE4 ~
##     aw               -0.678    0.363   -1.866    0.062   -1.139   -0.904
##     tm                0.040    0.212    0.191    0.849    0.075    0.059
##     ca               -0.186    0.462   -0.402    0.688   -0.271   -0.215
##     th                1.419    0.730    1.945    0.052    1.269    1.007
##   NE5 ~
##     NE3               0.174    0.416    0.418    0.676    0.174    0.261
##     NE4              -0.073    0.348   -0.210    0.834   -0.073   -0.085
##     aw               -0.256    0.618   -0.415    0.678   -0.431   -0.396
##     tm                0.053    0.245    0.216    0.829    0.098    0.090
##     ca                0.265    0.389    0.681    0.496    0.387    0.356
##     th                0.377    1.378    0.274    0.784    0.337    0.310
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
```

```
##   aw ~~
##     tm                1.863    0.420    4.434    0.000    0.600    0.600
##     ca                1.783    0.432    4.125    0.000    0.727    0.727
##     th                1.202    0.251    4.782    0.000    0.800    0.800
##   tm ~~
##     ca                2.046    0.422    4.852    0.000    0.759    0.759
##     th                0.784    0.217    3.608    0.000    0.474    0.474
##   ca ~~
##     th                0.932    0.216    4.323    0.000    0.714    0.714
##
## Intercepts:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               7.312    0.234   31.297    0.000    7.312    4.137
##    .AW2               7.221    0.240   30.059    0.000    7.221    3.726
##    .AW3               7.948    0.211   37.685    0.000    7.948    4.910
##    .AW4               7.065    0.238   29.685    0.000    7.065    3.674
##    .TM1               7.091    0.247   28.713    0.000    7.091    3.602
##    .TM2               6.909    0.284   24.302    0.000    6.909    3.277
##    .TM3               7.506    0.296   25.360    0.000    7.506    3.698
##    .TM4               7.156    0.236   30.377    0.000    7.156    4.157
##    .TM5               6.792    0.231   29.452    0.000    6.792    3.623
##    .CA2               7.234    0.246   29.401    0.000    7.234    3.808
##    .CA3               7.130    0.235   30.381    0.000    7.130    3.905
##    .CA4               7.468    0.207   36.024    0.000    7.468    4.815
##    .CA5               7.273    0.205   35.521    0.000    7.273    4.178
##    .TH1               7.571    0.174   43.578    0.000    7.571    5.409
##    .TH3               7.584    0.191   39.630    0.000    7.584    4.794
##    .TH4               7.455    0.203   36.769    0.000    7.455    4.659
##    .TH5               7.442    0.236   31.537    0.000    7.442    3.909
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3|t1           -1.348    0.453   -2.977    0.003   -1.348   -0.825
##     NE4|t1           -0.608    0.242   -2.514    0.012   -0.608   -0.482
##     NE5|t1            0.072    0.182    0.394    0.693    0.072    0.066
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##    .AW1               0.301    0.232    1.301    0.193    0.301    0.096
##    .AW2               1.885    0.362    5.209    0.000    1.885    0.502
##    .AW3               1.810    0.257    7.034    0.000    1.810    0.691
##    .AW4               1.415    0.319    4.441    0.000    1.415    0.383
##    .TM1               0.457    0.254    1.800    0.072    0.457    0.118
##    .TM2               0.681    0.201    3.391    0.001    0.681    0.153
##    .TM3               2.097    0.209   10.051    0.000    2.097    0.509
##    .TM4               0.845    0.172    4.907    0.000    0.845    0.285
##    .TM5               0.629    0.224    2.811    0.005    0.629    0.179
##    .CA2               1.479    0.175    8.473    0.000    1.479    0.410
##    .CA3               1.845    0.169   10.922    0.000    1.845    0.553
##    .CA4               0.996    0.143    6.966    0.000    0.996    0.414
##    .CA5               1.500    0.230    6.506    0.000    1.500    0.495
##    .TH1               1.160    0.132    8.768    0.000    1.160    0.592
##    .TH3               1.050    0.139    7.528    0.000    1.050    0.420
##    .TH4               1.362    0.249    5.468    0.000    1.362    0.532
```

```
##    .TH5               1.106    0.289    3.833    0.000    1.106    0.305
##    .NE3               1.000                               1.000    0.374
##    .NE4               1.000                               1.000    0.629
##    .NE5               1.000                               1.000    0.845
##     aw                2.823    0.641    4.403    0.000    1.000    1.000
##     tm                3.418    0.766    4.463    0.000    1.000    1.000
##     ca                2.128    0.546    3.901    0.000    1.000    1.000
##     th                0.800    0.239    3.346    0.001    1.000    1.000
##
## Scales y*:
##                    Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
##     NE3               0.612                               0.612    1.000
##     NE4               0.793                               0.793    1.000
##     NE5               0.919                               0.919    1.000
```